

MTH 202 Chapter 1: Three-Dimensional Space & Vectors

1.6 Planes in 3-Space Worksheet

1. Determine whether each of the following statements are true or false.

a) Two planes parallel to a third plane are parallel to each other.

This is true. Suppose the planes have normal vectors $\mathbf{n}_1$, $\mathbf{n}_2$, & $\mathbf{n}_3$. If the first two planes are parallel to the third, then this means that both $\mathbf{n}_1$ and $\mathbf{n}_2$ are scalar multiples of $\mathbf{n}_3$, that is for some constants c and d we have that $\mathbf{n}_1 = c\mathbf{n}_3$ and $\mathbf{n}_2 = d\mathbf{n}_3$, but this then implies that $\mathbf{n}_3 = \frac{1}{d}\mathbf{n}_2 \rightarrow \mathbf{n}_1 = \frac{c}{d}\mathbf{n}_2$ which shows that $\mathbf{n}_1$ & $\mathbf{n}_2$ are scalar multiples of each other and therefore the first two planes are also parallel.

b) Two lines perpendicular to a plane are parallel to each other.

This is true. If the lines are perpendicular to the plane then this means that the direction vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ and the normal $\mathbf{n}$ are parallel. This means that both $\mathbf{v}_1$ & $\mathbf{v}_2$ are parallel and so the lines are parallel.

c) Two planes that are parallel to a line are parallel.

This is false. If two planes are parallel to a line then this means that the normal vectors $\mathbf{n}_1$ & $\mathbf{n}_2$ are perpendicular to the direction vector, $\mathbf{v}$, of the line. However, we can easily create two non-parallel normal vectors that have a dot product with $\mathbf{v}$ that equals 0. For example, consider $\mathbf{n}_1 = \langle 1, 0, 0 \rangle$ & $\mathbf{n}_2 = \langle 0, 1, 0 \rangle$ and have $\mathbf{v} = \langle 0, 0, 1 \rangle$. We can clearly see that $\mathbf{n}_1$ & $\mathbf{n}_2$ are not scalar multiples of each other and therefore are not parallel.

d) Two planes that are perpendicular to a line are parallel.

This is true. If two planes are perpendicular to a line, then the normal vectors $\mathbf{n}_1$ & $\mathbf{n}_2$ must be parallel to the direction vector, $\mathbf{v}$, of the line. This means that both $\mathbf{n}_1$ & $\mathbf{n}_2$ are parallel and so the planes are parallel.

e) A plane and a line either intersect or are parallel.

This is true. This is equivalent to saying that the normal vector of the plane and the direction vector of the line are either perpendicular (meaning the plane and the line intersect) or that they are not perpendicular (the plane and line intersect). Clearly one of these must be true.

2. Find the equation of the plane.

a) The plane through the origin and perpendicular to the vector $\langle 1, -2, 5 \rangle$.

We already have a point on the plane of $(0, 0, 0)$ and we can use the given vector as the normal vector for the plane, so let $\mathbf{n} = \langle 1, -2, 5 \rangle$.

Therefore, the equation of the plane is

$$1(x-0) - 2(y-0) + 5(z-0) = 0 \rightarrow x - 2y + 5z = 0$$

b) The plane through the point $(5, 3, 5)$ and with normal vector $2\mathbf{i} + \mathbf{j} - \mathbf{k}$.

We already have a point on the plane of $(5, 3, 5)$ and we have a normal vector $\mathbf{n} = \langle 2, 1, -1 \rangle$.

Therefore, the equation of the plane is

$$2(x-5) + 1(y-3) - 1(z-5) = 0 \rightarrow 2x + y - z = 8$$

c) The plane through the point $(2, 0, 1)$ and perpendicular to the line $x = 3t$, $y = 2 - t$, $z = 3 + 4t$.

We already have a point on the plane of $(2, 0, 1)$. If we want the plane to be perpendicular to the given line then we need to have a normal vector that is parallel to the direction vector for the line. Therefore, we select $\mathbf{n} = \langle 3, -1, 4 \rangle$.

Therefore, the equation of the plane is

$$3(x-2) - (y-0) + 4(z-1) = 0 \rightarrow 3x - y + 4z = 10$$

d) The plane through the point $(1, -1, -1)$ and parallel to the plane $5x - y - z = 6$.

We already have a point on the plane of $(1, -1, -1)$. If we want the plane to be parallel to the given plane then we need to have a normal vector that is parallel to the normal vector for the given plane. Therefore, we select $\mathbf{n} = \langle 5, -1, -1 \rangle$.

Therefore, the equation of the plane is

$$5(x-1) - (y-(-1)) - (z-(-1)) = 0 \rightarrow 5x - y - z = 7$$

e) A plane through the point $(2, 4, 6)$ and perpendicular to the plane $z = x + y$.

We already have a point on the plane of $(2, 4, 6)$. If we want the plane to be perpendicular to the given plane then we need to have a normal vector that is perpendicular to the normal vector for the given plane $\langle 1, 1, -1 \rangle$. Therefore, we select $\mathbf{n} = \langle 0, -1, 1 \rangle$.

Therefore, the equation of the plane is

$$0(x-2) - (y-4) + 1(z-6) = 0 \rightarrow -y + z = 2$$

- f) The plane through the points $(0,1,1)$, $(1,0,1)$, & $(1,1,0)$.

We already have a point on the plane of $(0,1,1)$. Since all the given points are in the plane we will create two vectors from these points.

$$\text{Let } \mathbf{u} = \langle 1-0, 0-1, 1-1 \rangle = \langle 1, -1, 0 \rangle \text{ \& } \mathbf{v} = \langle 1-0, 1-1, 0-1 \rangle = \langle 1, 0, -1 \rangle.$$

We then set our normal vector to be $\mathbf{u} \times \mathbf{v}$ since this will create a vector that is perpendicular to both vectors and therefore perpendicular to the plane.

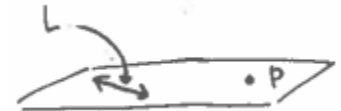
$$\mathbf{n} = \mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{vmatrix} = \langle 1-0, -(-1-0), 0-(-1) \rangle = \langle 1, 1, 1 \rangle$$

Therefore, the equation of the plane is

$$1(x-0) + 1(y-1) + 1(z-1) = 0 \rightarrow x + y + z = 2$$

- g) Find an equation of the plane that contains the line $L = \langle 1+t, 1, 2t \rangle$ and the point $(2, 4, 0)$.

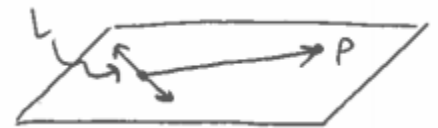
In order to construct the equation of a plane we need to have a point in the plane and a normal vector to the plane. We already have a point in the plane so we only need to find a normal vector to the plane.



The direction vector of L is $\vec{v} = \langle 1, 0, 2 \rangle$ is in the plane and we can create another vector from any point on L to the point P . We see that if $t = 0$ then on L we are at the point $(1, 1, 0)$. Therefore we can create a vector from $(1, 1, 0)$ to $(2, 4, 0)$ as $\vec{u} = \langle 1, 3, 0 \rangle$.

Therefore, a normal vector to the plane is given by

$$\vec{u} \times \vec{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 3 & 0 \\ 1 & 0 & 2 \end{vmatrix} = \langle 6-0, -(2-0), 0-3 \rangle = \langle 6, -2, -3 \rangle.$$

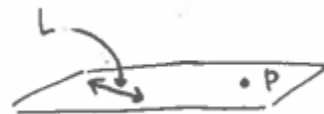


Therefore, the equation of the plane is

$$6(x-2) - 2(y-4) - 3(z-0) = 0$$

- h) Find an equation for the plane containing the point $(1, 2, 3)$ and the line L parameterized by $x = 1 + t$, $y = 2t$, & $z = 2 - t$.

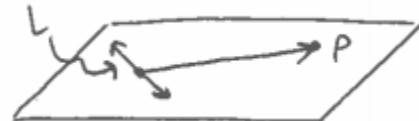
In order to construct the equation of a plane we need to have a point in the plane and a normal vector to the plane. We already have a point in the plane so we only need to find a normal vector to the plane.



The direction vector of L is $\vec{v} = \langle 1, 2, -1 \rangle$ is in the plane and we can create another vector from any point on L to the point P . We see that if $t = 0$ then on L we are at the point $(1, 0, 2)$. Therefore we can create a vector from $(1, 0, 2)$ to $(1, 2, 3)$ as $\vec{u} = \langle 0, 2, 1 \rangle$.

Therefore, a normal vector to the plane is given by

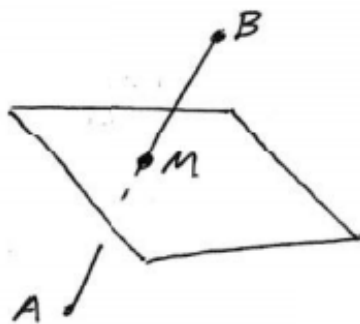
$$\vec{u} \times \vec{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 2 & 1 \\ 1 & 2 & -1 \end{vmatrix} = \langle -2 - 2, -(0 - 1), 0 - 2 \rangle = \langle -4, 1, -2 \rangle.$$



Therefore, the equation of the plane is

$$4(x - 1) - (y - 2) + 2(z - 3) = 0$$

- i) Find the equation of the plane P consisting of all points that are equidistance from $A = (2, 0, 1)$ and $B = (4, 2, 5)$.



We see that the point M located in P would be given by the midpoint between A and B . Therefore, we calculate $P = (3, 1, 3)$.

A normal vector to this plane could then be given by $\mathbf{n} = \langle 1, 1, 2 \rangle$.

Therefore, the equation of the plane is given by

$$1(x - 3) + 1(y - 1) + 2(z - 3) = 0$$

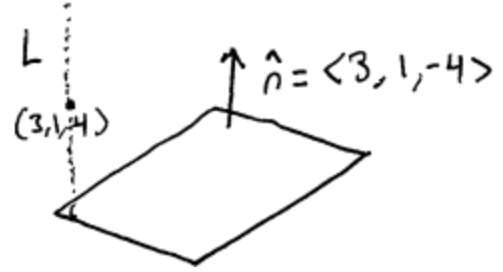
3. Determine parametric equations for the line which goes through the point $(3,1,4)$ and is perpendicular to the plane $3x+1y-4z=6$.

We see that the normal vector to the plane is given by $\mathbf{n} = \langle 3, 1, -4 \rangle$. If L is going to be perpendicular to the given plane then it must be parallel to $\mathbf{n}$. Therefore, $\mathbf{n} = \langle 3, 1, -4 \rangle$ serves as a direction vector for L .

So, parametric equations for the L are

$$L(t) = \langle 3, 1, 4 \rangle + \langle 3, 1, -4 \rangle t = \langle 3+3t, 1+t, 4-4t \rangle$$

$$\text{Or we could write this as } L(t) = \begin{cases} 3+3t \\ 1+t \\ 4-4t \end{cases}$$



4. Suppose that two planes have (non-zero) normal vectors $\mathbf{n}_1$ and $\mathbf{n}_2$ respectively, and that $\mathbf{n}_1 \times \mathbf{n}_2 = \mathbf{0}$.

Which of the following statements could possibly be true?

- A. The planes intersect in a line.
- B. The planes are orthogonal to each other.
- ☒ C. The planes are parallel to each other.
- ☒ D. The planes are equal to each other.

If $\mathbf{n}_1 \times \mathbf{n}_2 = \mathbf{0}$ this implies that the vectors are parallel. If the normal vectors of these planes are parallel, then the plane might be either parallel to each other or equal. Therefore, we assert that statements C & D are possibly true.

5. Find the angle $0 \leq \theta \leq \frac{\pi}{2}$ between the planes $z = x + \sqrt{2}y$ and $x - z = 5$. Express your answer in radians.

We can find the angle between two planes by finding the angle between the two normal vectors of the planes. We see that the normal vectors for these planes are $\mathbf{n}_1 = \langle 1, \sqrt{2}, -1 \rangle$ and $\mathbf{n}_2 = \langle 1, 0, -1 \rangle$. We calculate:

$$\cos(\theta) = \frac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{\|\mathbf{n}_1\| \|\mathbf{n}_2\|} = \frac{2}{2\sqrt{2}} \Rightarrow \theta = \cos^{-1}\left(\frac{1}{\sqrt{2}}\right) = \frac{\pi}{4}$$

6. Give a vector $\mathbf{v}$ perpendicular to the plane that contains the lines $x = 1 + t$, $y = 2 - 3t$, $z = 2 + 4t$ and $x = 2$, $y = -1 + 2t$, $z = 6 - t$.

If these two lines exist in the same plane then the direction vectors of both lines (say $\mathbf{a}$ and $\mathbf{b}$) exist in the same plane and we can find $\mathbf{v}$ by simply taking their cross-product.

We see that $\mathbf{a} = \langle 1, -3, 4 \rangle$ and $\mathbf{b} = \langle 0, 2, -1 \rangle$. Therefore,

$$\mathbf{v} = \mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -3 & 4 \\ 0 & 2 & -1 \end{vmatrix} = \langle 3 - 8, -(-1 - 0), 2 - 0 \rangle = \langle -5, 1, 2 \rangle$$

7. Which statement is true in $\mathbb{R}^3$?
- a) Two planes perpendicular to a third plane are parallel.
 - b) Two lines parallel to the same plane are parallel.
 - c) Two lines either intersect or are parallel.
 - ☒ d) Two planes either intersect or are parallel.

(a) is not always true. Consider the example of the xy -plane, xz -plane and yz -plane.

(b) Is not always true. Consider the example of the xy -plane and two intersecting lines, both with $z = 1$.

(c) This is not true. Lines can also be skew in 3-space.

(d) This state is *always* true.

Our answer is (d).

8. Consider the line L given by $x = 2 + 2t$, $y = 3$, $z = -1 + 2t$. Is L parallel to the plane $z - x - y = 2$?

We see that the direction vector of L is $\mathbf{v} = \langle 2, 0, 2 \rangle$ and the normal vector of the plane is $\mathbf{n} = \langle -1, -1, 1 \rangle$.

In order for the line and plane to be parallel it must be the case that $\mathbf{v}$ and $\mathbf{n}$ are perpendicular. We quickly observe that $\mathbf{v} \cdot \mathbf{n} = -2 + 0 + 2 = 0$ which mean that these vectors are perpendicular and therefore the line is parallel to the plane.

9. Let A be the plane given by $x - z = 1$ and B the plane given by $x + y + z = 2$.

a) Find a normal vector $\mathbf{n}$ for the plane A .

We can easily see from the equation of the plane A that $\mathbf{n} = \langle 1, 0, -1 \rangle$.

b) Find the angle between the two planes.

We note that the plane A has a normal vector $\mathbf{n} = \langle 1, 0, -1 \rangle$ and B has a normal vector $\mathbf{m} = \langle 1, 1, 1 \rangle$.

Since we can see that $\mathbf{n} \cdot \mathbf{m} = 0$ then we know that $\mathbf{n}$ and $\mathbf{m}$ are perpendicular and therefore the angle between the planes must be $\theta = \frac{\pi}{2}$.

c) Find the equation of a plane C which is perpendicular to both A and B .

We need a normal vector which is perpendicular to both $\mathbf{n}$ and $\mathbf{m}$. So, we take

$$\mathbf{v} = \mathbf{n} \times \mathbf{m} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & -1 \\ 1 & 1 & 1 \end{vmatrix} = \langle 0 - (-1), -(1 - (-1)), 1 - 0 \rangle = \langle 1, -2, 1 \rangle.$$

If we let our plane go through the point $(0, 0, 0)$ then the equation of our plane would be given by

$$1(x - 0) - 2(y - 0) + 1(z - 0) = 0.$$

10. Suppose that L is the line given by $x = 1 + 2t$, $y = -t$, $z = 2 + t$. Find the point P of the intersection of the line L with the plane whose equation is $3x - 2y + z = 14$.

If we are looking for the point of intersection between the line and the plane we can simply substitute out the values of x , y , and z in the following way.

$$\begin{aligned} 3x - 2y + z = 14 &\rightarrow 3(1 + 2t) - 2(-t) + (2 + t) = 14 \\ &\rightarrow 3 + 6t + 2t + 2 + t = 14 \\ &\rightarrow 5 + 9t = 14 \\ &\rightarrow t = 1 \end{aligned}$$

So the point of intersection occurs when $t = 1$. When $t = 1$ we see that the line is at the point $(1 + 2(1), -(1), 2 + (1)) = (3, -1, 3)$.

11. Let P be the plane with equation $x + 2z = 0$. Find the distance from the point $(-1, 3, 0)$ to the plane P.

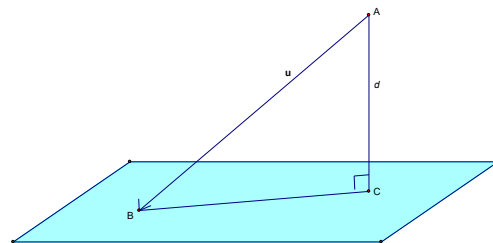
We see that the normal vector to this plane is $\mathbf{n} = \langle 1, 0, 2 \rangle$. We also note that the point $(2, 0, -1)$ is a point in the plane.

We create a vector $\mathbf{u}$ from $(-1, 3, 0)$ to $(2, 0, -1)$. Thus

$\mathbf{u} = \langle 3, -3, -1 \rangle$ We note that if we project $\mathbf{u}$ onto $\mathbf{n}$ and find the length of this projection, we will have found our distance.

Therefore, we compute:

$$d = |\text{comp}_{\mathbf{n}}(\mathbf{u})| = \left| \frac{\mathbf{u} \cdot \mathbf{n}}{\|\mathbf{n}\|} \right| = \left| \frac{3 + 0 - 2}{\sqrt{5}} \right| = \frac{1}{\sqrt{5}}$$



12. Determine the distance between the point $(-6, 3, 5)$ and the plane $x - 2y - 4z = 8$.

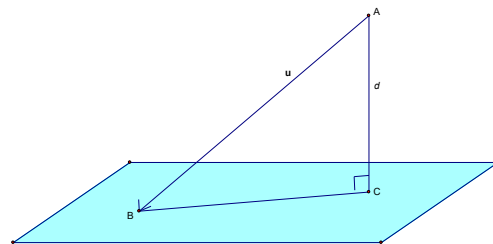
We see that the normal vector to this plane is $\mathbf{n} = \langle 1, -2, -4 \rangle$. We also note that the point $(0, 0, -2)$ is a point in the plane.

We create a vector $\mathbf{u}$ from $(-6, 3, 5)$ to $(0, 0, -2)$. Thus

$\mathbf{u} = \langle 6, -3, -7 \rangle$ We note that if we project $\mathbf{u}$ onto $\mathbf{n}$ and find the length of this projection, we will have found our distance.

Therefore, we compute:

$$d = |\text{comp}_{\mathbf{n}}(\mathbf{u})| = \left| \frac{\mathbf{u} \cdot \mathbf{n}}{\|\mathbf{n}\|} \right| = \left| \frac{6 + 6 + 28}{\sqrt{1 + 4 + 16}} \right| = \frac{40}{\sqrt{21}}$$



13. Find the distance between the two parallel planes $2x - 3y + z = 4$ and $4x - 6y + 2z = 3$.

We first select a point on the first plane $(2, 0, 0)$ and a point from the second plane $\left(0, -\frac{1}{2}, 0\right)$.

We see that the normal vector to the first plane is $\mathbf{n} = \langle 2, -3, 1 \rangle$.

We create a vector $\mathbf{u}$ from $\left(0, -\frac{1}{2}, 0\right)$ to $(2, 0, 0)$. Thus $\mathbf{u} = \left\langle 2, \frac{1}{2}, 0 \right\rangle$ We note that if we project $\mathbf{u}$ onto $\mathbf{n}$ and find the length of this projection, we will have found our distance.

Therefore, we compute:

$$d = |\text{comp}_{\mathbf{n}}(\mathbf{u})| = \left| \frac{\mathbf{u} \cdot \mathbf{n}}{\|\mathbf{n}\|} \right| = \left| \frac{0 + \frac{3}{2} + 0}{\sqrt{4 + 9 + 1}} \right| = \frac{3/2}{\sqrt{14}} = \frac{3}{2\sqrt{14}}$$

